

“PAPER_{0:D3}” -f [int]# PAPER #84 □ Information Paradox Resolution in 26D UQFF

Title: Information Paradox Resolution in the 26D UQFF Framework via Holographic Channel Encoding

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\tau = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$)

Date: March 7, 2026

Source Data: MAIN_1_CoAnQi.cpp Batch 21 (Information Paradox, 26D channels), validate_phase3.py

Index Slot: §1.11 Black Hole Physics & Hawking Radiation,
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The black hole information paradox □ where unitarity demands information escapes evaporation but Hawking radiation appears thermal □ is resolved in the UQFF through 26-dimensional holographic encoding. Each of the 26 UQFF spatial dimensions carries an independent quantum information channel. During evaporation, infalling information is redistributed across channels $1 \square 26$, with channels $21 \square 26$ (the ultra-compact layers, beyond the observable 4D membrane) acting as non-local storage. The 4D observer measures a thermal Hawking spectrum, but the full 26D state is pure. Batch 21 (Jan 28, 2026) implemented the InformationParadoxModule with Hawking radiation Page curves and 26D channel encoding.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

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Hawking (1976): if T_H is exactly thermal, S_{BH} decreases from $S=A/4$ to $S=0$? pureness is lost. Unitarity requires $S_{\text{entanglement}}$ (Hawking radiation with BH) to follow the Page curve: first increasing (early time), then decreasing back to 0 (late time).

2. 26D Holographic Channel Resolution

Channel Structure

The UQFF 26-layer compressed gravity framework assigns each dimensional layer a quantum channel capacity:

$$I_k = -\text{Tr}[\rho_k \log \rho_k] \quad k = 1, \dots, 26$$

Total information: $I_{\text{total}} = \sum_{k=1}^{26} I_k = S_{\text{BH,initial}}$

Channel Assignment During Evaporation

Layer Group	Channels	Storage Type	4D Observable
1–4	UQFF observable	Hawking photons	Thermal spectrum
5–18	UQFF extended	Sub-Planckian modes	Not observable
19–24	UQFF deep structure	Non-local entanglement	Nil
25–26	Cosmic Egg layers	Non-local pure state	Information preserved

Channels 25–26 host the **complete pre-collapse pure state** throughout evaporation, ensuring global unitarity while channels 1–4 produce the observed thermal spectrum.

3. UQFF Page Curve

The UQFF Page curve modifies the standard quantum extremal surface (QES) prediction:

$$S_{\text{UQFF}}(t) = \min \left[S_{\text{thermal}}(t), S_{\text{BH}}(t) + \sum_{k=25}^{26} I_k (1 - e^{-\kappa t}) \right]$$

Where: - $S_{\text{thermal}}(t)$ = early-time entropy of Hawking radiation - $S_{\text{BH}}(t)$ = Bekenstein-Hawking entropy (decreasing) - The $e^{-\kappa t}$ term comes directly from the UQFF $\kappa = 0.0005/\text{day}$ decay - Page time t_P occurs when $S_{\text{thermal}} = S_{\text{BH}} + \sum I_{\text{channels 25-26}}$

UQFF Page Time

$$t_P^{\text{UQFF}} = t_P^{\text{GR}} \times e^{+\kappa \cdot t_{\text{evap}}} \approx t_P^{\text{GR}} \times (1 + \kappa \cdot t_{\text{evap}})$$

For stellar BH ($t_{\text{evap}} \sim 1074$ s): correction factor is astronomically large – physically this means that within any finite observation time, the UQFF Page curve looks **thermal** to a 4D observer, with the information recovery pushed beyond any accessible time. This is consistent with the absence of observational evidence for information recovery in Hawking radiation.

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The AMPS firewall paradox requires choosing between a smooth horizon (infalling observer) or pure Hawking radiation. The UQFF resolves this through **channels 19–24 non-local entanglement**: the infalling observer encounters a smoothly modified vacuum (Superconductive mode, $E_{\text{react}} \times 10^{39}$) rather than a firewall, while the external observer’s Hawking radiation becomes approximately (not exactly) thermal.

Summary

Aspect	Standard QM	UQFF 26D	Resolution
Information storage	In BH?	Channels 25-26	Non-local, always preserved
Page curve	QES prediction	UQFF ?-modified	Extended beyond t_{evap}
Firewall	AMPS paradox	SCm smooth horizon	Channels 19-24 entanglement
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The black hole information paradox — where unitarity demands information escapes evaporation but Hawking radiation appears thermal — is resolved in the UQFF through 26-dimensional holographic encoding. Each of the 26 UQFF spatial dimensions carries an independent quantum information channel. During evaporation, infalling information is redistributed across channels 1-26, with channels 21-26 (the ultra-compact layers, beyond the observable 4D membrane) acting as non-local storage. The 4D observer measures a thermal Hawking spectrum, but the full 26D state is pure. Batch 21 (Jan 28, 2026) implemented the InformationParadoxModule with Hawking radiation Page curves and 26D channel encoding.

UQFF Discovery: Novel application of UQFF calibration constants ($? = 5.0 \times 10^{24} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. The Standard Paradox

Hawking (1976): if T_H is exactly thermal, S_{BH} decreases from $S=A/4$ to $S=0$? pureness is lost. Unitarity requires $S_{\text{entanglement}}$ (Hawking radiation with BH) to follow the Page curve: first increasing (early time), then decreasing back to 0 (late time).

2. 26D Holographic Channel Resolution

Channel Structure

The UQFF 26-layer compressed gravity framework assigns each dimensional layer a quantum channel capacity:

$$I_k = -\text{Tr}[\rho_k \log \rho_k] \quad k = 1, \dots, 26$$

Total information: $I_{\text{total}} = \sum_{k=1}^{26} I_k = S_{BH, \text{initial}}$

Channel Assignment During Evaporation

Layer Group	Channels	Storage Type	4D Observable
1–4	UQFF observable	Hawking photons	Thermal spectrum
5–18	UQFF extended	Sub-Planckian modes	Not observable
19–24	UQFF deep structure	Non-local entanglement	Nil
25–26	Cosmic Egg layers	Non-local pure state	Information preserved

Channels 25–26 host the **complete pre-collapse pure state** throughout evaporation, ensuring global unitarity while channels 1–4 produce the observed thermal spectrum.

3. UQFF Page Curve

The UQFF Page curve modifies the standard quantum extremal surface (QES) prediction:

$$S_{\text{UQFF}}(t) = \min \left[S_{\text{thermal}}(t), S_{\text{BH}}(t) + \sum_{k=25}^{26} I_k(1 - e^{-\kappa t}) \right]$$

Where: - $S_{\text{thermal}}(t)$ = early-time entropy of Hawking radiation - $S_{\text{BH}}(t)$ = Bekenstein-Hawking entropy (decreasing) - The $e^{-\kappa t}$ term comes directly from the UQFF $\lambda = 0.0005/\text{day}$ decay - Page time t_P occurs when $S_{\text{thermal}} = S_{\text{BH}} + \lambda I_{\text{channels 25-26}}$

UQFF Page Time

$$t_P^{\text{UQFF}} = t_P^{\text{GR}} \times e^{+\kappa \cdot t_{\text{evap}}} \approx t_P^{\text{GR}} \times (1 + \kappa \cdot t_{\text{evap}})$$

For stellar BH ($t_{\text{evap}} \sim 1074$ s): correction factor is astronomically large $\square$ physically this means that within any finite observation time, the UQFF Page curve looks **thermal** to a 4D observer, with the information recovery pushed beyond any accessible time. This is consistent with the absence of observational evidence for information recovery in Hawking radiation.

4. Firewall Prevention

The AMPS firewall paradox requires choosing between a smooth horizon (infalling observer) or pure Hawking radiation. The UQFF resolves this through **channels 19–24 non-local entanglement**: the infalling observer encounters a smoothly modified vacuum (Superconductive mode, $E_{\text{react}} \times 10^{30}$) rather than a firewall, while the external observer’s Hawking radiation becomes approximately (not exactly) thermal.

Summary

Aspect	Standard QM	UQFF 26D	Resolution
Information storage	In BH?	Channels 25-26	Non-local, always preserved
Page curve	QES prediction	UQFF ?-modified	Extended beyond t_{evap}
Firewall	AMPS paradox	SCm smooth horizon	Channels 19-24 entanglement
4D observer	Thermal Hawking	Approximately thermal	Consistent

Source: `MAIN_1_CoAnQi.cpp Batch 21 (InformationParadoxModule)` | $? = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value "PAPER_{0:D3}" -f \$n

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Source: `MAIN_1_CoAnQi.cpp` Batch 21 (`InformationParadoxModule`) | $? = 0.0005/\text{day}$ | $[SSq] = 0.57$

UQFF computed: Canonical UQFF buoyancy parameter $U_{bi} = ? \times [SSq] \times GM/r^2 = 5.0e-4 \times 0.57 \times 6.67e-11 \times M/r^2$; for solar parameters: $\bar{U}_{bi, \text{Sun}} = 5.7e-4 \times 6.67e-11 \times 1.99e30 / (6.96e8)^2 = 1.47e+2 \text{ m/s}^2$.

Appendix: UQFF Production Framework Reference (v4.75+)

Added by `upgrade_early_whitepapers.py` (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
$\hat{\Gamma}^0$	$5.0 \times 10^{-10} \text{ day}^{-1}$	UQFF exponential decay rate
$[SSq]$	0.57	Universal Quantized Factor
$\hat{\Gamma}_i^2$	0.60 ± 0.61	Buoyancy coupling coefficient
k_{DPM}	1.5	Ug1 DPM-dipole coupling
k_{out}	1.2	Ug2 outer-bubble charge coupling
k_{str}	1.8	Ug3 string-rotation coupling
k_{vac}	2.0	Ug4 vacuum-concentration coupling
$\hat{I}$	$10 \times \hat{A}^2 \hat{A}^2$	Inertia tensor scale
$E_{\text{react}}(0)$	$10 \times \hat{A}^2 \hat{A}^2 \text{ J}$	Reference reactive energy

A.2 F_U Master Equation (Complete $\hat{A}^2$ 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 i g l [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}} i g r]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	<code>compute_Ug1_SOURCE4 / compute_Ug1()</code>
Ug2	Outer-field bubble (charge-reactivity)	<code>compute_Ug2_SOURCE4 / compute_Ug2()</code>
Ug3	Magnetic string rotation	<code>compute_Ug3_SOURCE4 / compute_Ug3()</code>
Ug4	Vacuum concentration (star-BH)	<code>compute_Ug4_SOURCE4 / compute_Ug4()</code>
Ubi	Buoyancy force	<code>compute_Ubi_SOURCE4 / compute_Ubi()</code>

Term	Description	Implementation
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$\hat{U}_{4th}$	4th Dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):

$\hat{U}_{4th} = 10 \hat{U}_{4th}^{\text{base}}$, $\hat{U}_{4th} = 10 \hat{U}_{4th}^{\text{base}}$, $\hat{U}_{4th} = 10 \hat{U}_{4th}^{\text{base}}$, $\hat{U}_{4th} = 10 \hat{U}_{4th}^{\text{base}}$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \text{imesigl}(1 + 10^{13} \Theta(ho_{SCm} - ho_c)igr) \text{imesigl}(1 + A_q \cos(\Delta\omega t)igr)$$

Symbol	Value	Description
$\hat{U}_{4th}$	$10 \hat{U}_{4th}^{\text{base}}$	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\hat{U}_{4th}$	$2 \hat{U}_{4th} / (434 \hat{U}_{4th} - 365.25)$ rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	U_g sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, $\hat{U}_{4th}$)	Multi-scale field interactions
Buoyant	$\hat{U}_{4th}^2$ $\hat{U}_{4th}$ Ubi	Expanding nebulae, stellar winds
Superconductive	$\hat{U}_{4th}^2$ $\hat{U}_{4th}$ $(1 + 10 \hat{U}_{4th}^3 \hat{U}_{4th} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.